Sahar Darafsh



Software and Machine Learning Engineer · Applied AI

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SUMMARY

Software and machine learning engineer who ships LLM features from idea to production, with an MSc in Artificial Intelligence and Robotics and 4.5 years of production backend work in Munich. I build LLM workflows on self-hosted open-source models with structured outputs, validation and human review, and evaluate models on my own labelled data for accuracy, latency and GPU use. Full stack in Python, TypeScript and Java. I use Claude Code daily, including custom agents that write and review unit tests.

SKILLS

NLP & LLMs: Hugging Face Transformers · XLM-RoBERTa · zero-shot classification (NLI) · LLM APIs (OpenAI-compatible API, Ollama, Qwen 2.5) · JSON output from LLMs · prompt engineering · sentiment analysis · multilingual text (English, Persian)

ML / Deep Learning: Keras · TensorFlow · PyTorch · CNN training · fine-tuning · evaluation (accuracy, F1, confusion matrix) · Hugging Face Hub

Computer Vision: OpenCV · video/motion-based feature encoding · image classification · activity recognition · background subtraction · embedded CV design (Xilinx ZCU102, thesis)

Python & Tooling: Python · pandas · Git · Claude Code · pytest · ruff · CLI design · layered architecture · automated unit testing

Backend, MLOps & DevOps: Java · TypeScript · NestJS · REST APIs · gRPC · Protobuf (Buf) · MLflow · Spring Framework · Next.js · React · Docker · Kubernetes · CUDA · AWS · CI/CD (Jenkins, GitHub Actions, GitLab CI) · SonarQube · SQL (MySQL, PostgreSQL) · TDD/JUnit

WORK EXPERIENCE

Full Stack Developer · Convien GmbH, Munich | Mar 2025 – Oct 2026

- Took features from requirements to production across the Wicket UI and Java backend of a large-scale meeting-place optimizer, with sub-second responses.
- Used Claude Code daily on production code and built two agents to write and review unit tests, raising coverage from 0% to **95–98%** with a final team review.
- Introduced SonarQube and drove **1000+** critical issues down to zero.

Software Developer · TESTIFI GmbH, Munich (Remote) | Mar 2022 – Mar 2025

- Employee of the Month (Mar 2024) for fast, high-quality bug fixes.
- Turned stakeholder requirements into core features of QualityOps, a Java tool for evaluating software quality, using TDD.
- Shipped a Flutter Android app; contributed to CAST, a cross-team test-automation initiative.

Deep Learning Mentor and Teaching Assistant · Maktabkhooneh, Tehran (Remote) | Jun 2021 – Mar 2022

- Taught and mentored **100+** students in deep learning, CNNs and reinforcement learning.
- Co-developed the curriculum with industry examples.

Machine Learning Engineer · danaXa, Canberra (Remote) | Jan 2021 – Mar 2022

- Built a resource-efficient computer vision pipeline with YOLO on a Jetson Nano.
- Delivered low-latency, on-device video analytics for occupancy monitoring.

AI/ML Researcher and Project Reviewer · AUMS, Abadan (Remote) | Jun 2020 – Mar 2022

- Reviewed deep-learning projects in medical image analysis.
- Evaluated each project's model architecture, training and metrics.

RESEARCH & PROJECTS

Daily Journal Sentiment Analysis (private GitLab repos)

Medium write-up

- Classifies English and Persian journal entries as good, neutral or bad days, with a reason.
- Evaluated a zero-shot model, a sentiment model and a local LLM on my own labels for accuracy, latency and GPU use.
- Served over gRPC, tracked in MLflow, released automatically.

LLM Listing Assistant

github.com/sdrfsh/ev-listing-auditor · Live demo

- Full stack LLM app: a self-hosted LLM (Qwen 2.5 on Ollama, OpenAI-compatible API) extracts sellers' chat messages into structured JSON edits.
- Validates every output with regex VIN extraction, an allowed-field schema and human approval of each edit before a listing changes.
- NestJS/TypeScript, PostgreSQL, Next.js/React; Docker, GitHub Actions, Kubernetes.

Real-Time Activity & Intention Recognition: MSc thesis to open-source system

github.com/sdrfsh/realtime-activity-and-intention-recognition · huggingface.co/sdrfsh/alexnet-door-entry-classifier · arxiv.org/abs/2107.12744 · Medium write-up

- Video-to-decision pipeline that compresses motion clips into one image for a lightweight CNN, designed for an embedded FPGA. First-author arXiv paper on a dataset I built (**98.78%**).
- Rebuilt in 2026 as a tested Python CLI; the Hugging Face model reaches **98.84%** accuracy (**F1 0.99**) on held-out data.

adaptive-motion-preprocessing: open-source Python package

github.com/sdrfsh/adaptive-motion-preprocessing · pypi.org/project/adaptive-motion-preprocessing · Medium write-up

- Turns video frames into motion images for neural networks with pluggable preprocessing stages. Released on PyPI, with CI tests on Linux, Windows and macOS.

EDUCATION

MSc, Artificial Intelligence and Robotics · Amirkabir University of Technology, Tehran | Sep 2017 – Jun 2020

Thesis: "Designing an Embedded System for Activity Recognition Using a Surveillance Camera" (see Research & Projects for the resulting repo, paper, and model)

BSc, Computer Engineering · Isfahan University of Technology, Isfahan | Sep 2011 – Sep 2016

CERTIFICATES

- AI for Medicine Specialization, DeepLearning.AI
- Deep Learning Specialization, DeepLearning.AI
- Reinforcement Learning Specialization, University of Alberta
- Generative Adversarial Networks (GAN) Specialization, DeepLearning.AI

LANGUAGES

English (Fluent) · German (B1 telc certificate, B2 in progress) · Persian (Native)